

Functional genetic report · longevity

Heart Failure and Cardiomyopathies

Sérgio F · Report date: 03-10-2026

What the genes say about the strength and structure of the cardiac muscle, the tendency to fibrosis and remodeling, and how telmisartan and dapagliflozin act in this pathway. It complements the heart failure section of the advanced cardiovascular report. Only genes (chip + local imputation + polygenic scores).

Contents

• Summary	3
• How to read this report	4
• Polygenic scores	5
• Heart failure and dilated cardiomyopathy	6
• Hypertrophic cardiomyopathy and left ventricular mass	7
• Fibrosis and remodeling: renin-angiotensin-aldosterone axis and telmisartan	8
• Natriuretic peptides (BNP, ANP)	9
• Mechanisms of ageing	10
• Practical suggestions	11
• Limitations	12
• Methods and sources	13

Summary

13

SNPs analyzed

10

in the chip

3

imputed ($DR2 \geq 0.8$)

0

SNP not found in the data

8

Polygenic scores

Heart failure is the final route of many myocardial aggressions. Primary cardiomyopathies have their own genetics: dilated (large and weak ventriculum) and hypertrophic (thick and rigid wall) depend on rare variants of sarcomere (TTN, MYH7, MYBPC3), but also on many common loci that change risk and severity. Myocardial fibrosis, the common substrate of cardiac aging and insufficiency with preserved ejection fraction, is promoted by angiotensin II (via AT1 receptor) and aldosterone, and blocked by natriuretic peptides.

In summary: dilated cardiomyopathy with very low genetic risk (714 thousand variant score at the 2nd percentile) and low global heart failure (the 22nd percentile), despite two unfavourable loci (ZBTB17 CC, CDKN1A GA). Hypertrophic cardiomyopathy has a score above average (the 76th percentile in the large score, 89 in the small one), without the risk allele of SMARCB1 or MYBPC3. The renin-angiotensin axis is intermediate (AGT Met235Thr, ACE I/D and AGTR1 1166 in heterozygous); telmisartan blocks the AT1 receptor, the final step of this profibrotic pathway, whatever the genotype.

Favourable

- **Dilated cardiomyopathy at the 2nd percentile** (714 thousand variants score) and heart failure at the 22nd percentile; mean genetic ejection fraction (the 51st percentile); TTR without V142I (cardiovascular report).
- **SMARCB1 rs5760054 TT** and **MYBPC3 rs3729989 TT**: without alleles of hypertrophic cardiomyopathy and more ventricular mass; mean genetic left ventricular mass (the 56th percentile).
- **Telmisartan and dapagliflozin already in the plan**: act directly in the remodeling (AT1 block and less fibrosis), which makes the intermediate genotype of the renin-angiotensin axis little relevant.

Unfavourable

- **Hypertrophic cardiomyopathy above average**: 374 thousand variants score at the 76th percentile and 27 variants at the 89th percentile; FHOD3 rs2644262 TC.
- **ZBTB17 rs10927875 CC** (common allele of cardiomyopathy dilated in homozygous) and **CDKN1A rs3176326 GA** (p21, linked to senescence; risk of both cardiomyopathies). Small score of reduced ejection fraction at the 92nd percentile.
- **Renin-angiotensin intermediate axis**: AGT Met235Thr AG, ACE I/D heterozygous (rs4343 GA) and AGTR1 1166 AC NPPB and NPPA without the highest natriuretic peptide alleles (NT-proBNP genetic at the 23rd percentile).

Interpretation: low risk profile for heart failure and dilated cardiomyopathy. The point to be monitored is the genetic tendency for hypertrophy and fibrosis, which a baseline echocardiogram clarifies and that telmisartan, dapagliflozin, exercise and the absence of alcohol already contradict.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 10 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The scores of dilated cardiomyopathy (714 thousand variants), hypertrophic (374 thousand and 27), ejection fraction (266), reduced ejection fraction (22), left ventricular mass (465) and NT-proBNP (554 thousand) were calculated in this report; heart failure (39 variants) was reused from the advanced cardiovascular report. Large scores are more robust than small ones when they diverge.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Heart failure (39 variants)	PGS003969	38 of 39 (97%)	p22	Typical / no judgement
Dilated cardiomyopathy (714 thousand variants)	PGS004861	698 100 of 713 932 (98%)	p2	Favourable
Left ventricular ejection fraction (266 variants)	PGS001412	214 of 265 (81%)	p51	Typical / no judgement
Reduced ejection fraction (22 variants)	PGS002279	21 of 22 (96%)	p92	Unfavourable
Hypertrophic cardiomyopathy (374 thousand variants)	PGS004910	365 561 of 374 190 (98%)	p76	Typical / no judgement
Hypertrophic cardiomyopathy (27 variants)	PGS000739	26 of 27 (96%)	p89	Unfavourable
Left ventricular mass index (465 variants)	PGS003427	400 of 465 (86%)	p56	Typical / no judgement
NT-proBNP in blood (554 thousand variants)	PGS000270	543 654 of 554 137 (98%)	p23	Typical / no judgement

Heart failure and dilated cardiomyopathy

Dilated cardiomyopathy has an identifiable genetic cause in about 30–40% of cases (especially truncated TTN), but recent GWAS showed that common loci explain a large part of the remaining heritability (BAG3, HSPB7/ ZBTB17, CDKN1A).

Polygenic scores of this theme: Heart failure (39 variants) (PGS003969): the 22nd percentile

Typical / no judgement; Dilated cardiomyopathy (714 thousand variants) (PGS004861): the 2nd percentile

Favourable; Left ventricular ejection fraction (266 variants) (PGS001412): the 51st percentile

Typical / no judgement; Reduced ejection fraction (22 variants) (PGS002279): the 92nd percentile **Unfavourable**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
ZBTB17	rs10927875	CC	chip	C	2	0.69	ZBTB17 (1p36, near HSPB7): C allele (common) increases dilated cardiomyopathy (OR ~1.14). (strong evidence)	Unfavourable
CDKN1A	rs3176326	GA	chip	A	1	0.18	CDKN1A (p21, the cell cycle brake linked to senescence): allele A increases hypertrophic and dilated cardiomyopathy. (strong evidence)	Unfavourable
LSM3	rs17226476	GG	chip	A	0	0.23	LSM3 (RNA processing): A allele, less common, increases dilated cardiomyopathy. (moderate evidence)	Typical / no judgement
CCDC136	rs17165191	TT	chip	C	0	0.14	CCDC136/FLNC (filamine C): C allele, less common, increases dilated cardiomyopathy. (moderate evidence)	Typical / no judgement

Favourable

- Dilated cardiomyopathy score at the 2nd percentile
- Heart failure (39 variants) at the 22nd percentile
- LSM3 GG and CCDC136 TT (without the least common risk alleles)
- Ejection fraction (266 variants) at the 51st percentile

Unfavourable

- ZBTB17 rs10927875 CC
- CDKN1A rs3176326 GA
- Reduced ejection fraction (22 variants) at the 92nd percentile

Interpretation: The large score (714 thousand variants) is the most informative and is very low. The small score of reduced ejection fraction (22 variants) points in the opposite direction, but it is much less robust. CDKN1A encodes p21, one of the effectors of cellular senescence; rapamycin reduces senescence and improves cardiac function in elderly rats in experimental studies.

Hypertrophic cardiomyopathy and left ventricular mass

Hypertrophic cardiomyopathy affects about 1 in 500 people; half has a sarcomere variant and the other half has a polygenic risk (FHOD3, SMARCB1, BAG3, CDKN1A).

Polygenic scores of this theme: Hypertrophic cardiomyopathy (374 thousand variants) (PGS004910): the 76th percentile **Typical / no judgement**; Hypertrophic cardiomyopathy (27 variants) (PGS000739): the 89th percentile **Unfavourable**; Left ventricular mass index (465 variants) (PGS003427): the 56th percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
FHOD3	rs2644262	TC	imputation DR2=0.97	C	1	0.27	FHOD3 (sarcomere assembly): C allele increases hypertrophic cardiomyopathy. (strong evidence)	Typical / no judgement
SMARCB1	rs5760054	TT	imputation DR2=1.00	C	0	0.19	SMARCB1 (chromatin remodeling): C allele increases hypertrophic cardiomyopathy. (strong evidence)	Favourable
MYBPC3	rs3729989	TT	chip	C	0	0.14	Myosin-binding protein C (the most frequent gene of familial hypertrophic cardiomyopathy): this common polymorphism changes the mass of the left ventricle; the least common allele C increases the mass. (moderate evidence)	Typical / no judgement

Favourable

- SMARCB1 rs5760054 TT
- MYBPC3 rs3729989 TT
- Left ventricular mass (465 variants) at the 56th percentile

Unfavourable

- 374 thousand variants score at the 76th percentile
- 27 variant score at the 89th percentile
- FHOD3 rs2644262 TC

Interpretation: Polygenic risk of hypertrophy above average, without strong isolated variants. The polygenic risk of hypertrophic cardiomyopathy is also associated with a thicker wall and more rigidity of the ventricle in the general population. The BAG3 rs2234962 variant of the cardiovascular report is neutral.

Fibrosis and remodeling: renin-angiotensin-aldosterone axis and telmisartan

Angiotensin II, through the AT1 receptor, activates TGF- β , fibroblasts and collagen production in the myocardium; aldosterone (CYP11B2) amplifies fibrosis. Telmisartan blocks AT1 and, unlike ECA inhibitors, it does not depend on the ECA genotype; it also has partial action on PPAR γ .

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
AGT Met235Thr	rs699	AG	chip	G	1	0.41	Angiotensinogen: the Thr235 form (G allele) increases angiotensinogen; more angiotensin II means more pro-fibrotic signal (TGF- β) in the myocardium, which telmisartan blocks in the AT1 receptor. (moderate evidence)	Typical / no judgement
ACE I/D (2350A>G)	rs4343	GA	chip	G	1	0.56	ECA: the G allele marks the deletion (D), with more ECA and more angiotensin II; associated with hypertrophy and fibrosis of the left ventricle. Telmisartan acts after ECA and is independent of this genotype. (moderate evidence)	Typical / no judgement
AGTR1 A1166C	rs5186	AC	chip	C	1	0.28	AT1 receptor, the target of telmisartan: C allele is associated with more AT1 signal and more fibrosis; in some studies, a greater response to AT1 blockers. (weak evidence)	Typical / no judgement
CYP11B2 -344C>T	rs1799998	AA	chip	—	—	—	Aldosterone-synthase: the -344 variant changes aldosterone production, which promotes fibrosis in the myocardium; telmisartan indirectly reduces aldosterone. (weak evidence)	Typical / no judgement

Favourable

- Telmisartan in use (AT1 block)

Unfavourable

- AGT Met235Thr AG (more angiotensinogen, 1 copy)
- ACE rs4343 GA (I/D heterozygote)
- AGTR1 A1166C AC

Interpretation: Intermediate genotype in all steps of the axis: neither the form of low activity, nor the one of high activity in homozygous. AGTR1 1166C is associated in some studies with more AT1 signal and a greater response to AT1 blockers. The CYP11B2 -344 AA (strand +) is information. For fibrosis, it is essential that the pathway is blocked by telmisartan; dapagliflozin reduces fibrosis and remodeling by another route.

Natriuretic peptides (BNP, ANP)

ANP and BNP are produced by the heart when its walls are stretched; they stop fibrosis, hypertrophy and sodium retention. The alleles that increase these peptides are protective, but make the NT-proBNP rise from the analyses.

Polygenic scores of this theme: NT-proBNP in blood (554 thousand variants) (PGS000270): the 23rd percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
NPPB	rs198389	AA	chip	G	0	0.41	BNP promoter: G allele increases BNP. Natriuretic peptides stop fibrosis and hypertrophy; more base peptide is a protective trait, but rises the value of NT-proBNP in the analyzes. (moderate evidence)	Typical / no judgement
NPPA	rs5068	AA	imputation DR2=1.00	G	0	0.07	NPPA: the G allele, less common, increases the circulating ANP and BNP (protective trait). (moderate evidence)	Typical / no judgement

Favourable

- NT-proBNP at the 23rd percentile (lower base values, which facilitates the analysis)

Unfavourable

- NPPB rs198389 AA and NPPA rs5068 AA (without the highest peptide alleles)

Interpretation: Genetically low base natriuretic peptides: less base natriuretic protection, but an easier NT-proBNP to interpret. A value of NT-proBNP to rise in the analysis has, therefore, more significance.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Remodelling of the extracellular matrix	AGT AG; ACE I/D; AGTR1 AC	Telmisartan; dapagliflozin	Intermediate route, blocked by drugs.
Stiffness of the Extracellular Matrix	Hypertrophic cardiomyopathy p76-89	MYBPC3 TT; ventricular mass p56	Tendency for stiffness to be monitored.
Cell Senescence	CDKN1A GA	Rapamycin in use	Mixed.
Proteostasis loss	ZBTB17 CC; FHOD3 TC	Dilated cardiomyopathy p2; SMARCB1 TT	Globally favourable.
Endocrine dysregulation	NPPB AA; NPPA AA	—	Base peptides are a little low.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Baseline echocardiogram	The same echocardiogram suggested in the aorta report should measure the thickness of the walls, the mass and diastolic function of the left ventricle; it is the way to confirm or remove the genetic tendency for hypertrophy. <i>suggestion (not applied)</i>	Hypertrophic cardiomyopathy p76-89
NT-proBNP in the analysis	Include NT-proBNP (and high sensitivity troponin, if the doctor agrees) once to have a baseline value; with low genetic NT-proBNP, an increase has more meaning. <i>suggestion (not applied)</i>	NT-proBNP p23
Telmisartan and dapagliflozin	Maintain as prescribed: are the two drugs of the plan with evidence against fibrosis and remodeling (dapagliflozin reduced hospitalizations due to heart failure in large trials). <i>suggestion (not applied)</i>	Renin-angiotensin intermediate axis
Exercise	Aerobic training and regular strength, which improves ventricular compliance; avoid extreme volumes of resistance during years without follow-up, which are associated with fibrosis in athletes. <i>suggestion (not applied)</i>	Hypertrophic cardiomyopathy p76-89
Sleep apnea	If there is strong snoring or non-repairing sleep, do a sleep study: untreated apnea accelerates hypertrophy and fibrosis. <i>suggestion (not applied)</i>	Hypertrophy
Coordination with the plan	Without alcohol (alcoholic cardiomyopathy). Ferritin below 100. Omega-3 3.2-4 g/day (already taking fish oil). D3 10 000 IU. Melatonin 0.3-1 mg. Curcumin, luteolin, apigenin and quercetin in pause while ALT is high. <i>suggestion (not applied)</i>	V2 plan rules

Limitations

- The rare variants that cause cardiomyopathies (TTN truncated, MYH7, MYBPC3, LMNA, FLNC, DSP) are not read reliably by the chip.
- The genes of the renin-angiotensin axis have small and inconsistent effects between studies; the interpretation here is limited to fibrosis and response to telmisartan.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use echocardiogram or analysis.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population ($n=633$); effect alleles confirmed by the expected frequency.
- Scores: PGS004861 (dilated cardiomyopathy), PGS004910 and PGS000739 (hypertrophic cardiomyopathy), PGS001412 and PGS002279 (ejection fraction), PGS003427 (left ventricle mass) and PGS000270 (NT-proBNP) calculated in this report; PGS003969 reused from the advanced cardiovascular report.
- Main references: Tadros 2021 and Baxter 2021 (GWAS of hypertrophic cardiomyopathy); Garnier 2021 and Jurgens 2024 (GWAS of dilated cardiomyopathy); Shah 2020 (GWAS heart failure); Newton-Cheh 2009 (NPPA/NPPB); Rigat 1990 (ACE I/D); Jeunemaitre 1992 (AGT Met235Thr); McMurray 2019 (dapagliflozin in heart failure); Dai 2014 (rapamycin and cardiac aging in rats).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).